

Joseph Farkas

(770) 990-5495 | josephfarkas@gatech.edu | [linkedin.com/in/farkasjoseph](https://www.linkedin.com/in/farkasjoseph) | github.com/farkasjoseph | US Citizen

EDUCATION

Georgia Institute of Technology

Atlanta, GA

Bachelor of Science in Computer Science, Devices and Intelligence

Expected Dec 2027

Minors in Robotics (Aerospace Systems) and German; Concert Orchestra Violinist; BS/MS Applicant

GPA: 4.00

- Relevant Coursework: Data Structures, Algorithm Design and Analysis, Robotics and Perception

PROFESSIONAL EXPERIENCE

Autonomous Systems Lead, Software Mentor

Oct 2021 – Present

North Gwinnett Robotics

Suwanee, GA

- Mentoring 15 high school students in sensor fusion and control using C++ for 115lb competition robot
- Integrating computer vision localization with on-the-fly A* pathfinding to power fully autonomous scoring
- World Comp. Autonomous Award & Division Champions. **Rated #1 globally for autonomous performance**

Research Assistant

Jan 2026 – Present

Georgia Tech Engineering Space Policy Lab

Atlanta, GA

- Modeling the downrange risk of space launches to inform policy decisions on launch safety and risk mitigation
- Scraping FAA NOTAMs to build a dataset of launch attempts; Plotting risk contours using QGIS

Software Intern

Aug 2024 – May 2025

Viasat Inc.

Duluth, GA

- Developed C/C++ firmware and Typescript frontend for prototype three-axis satellite antennas
- Built algorithms for analyzing signal quality and atmospheric modeling; integrated CI pipelines for validation

UX Assistant

Aug 2025 – Present

Georgia Tech PACE High Performance Computing Center

Atlanta, GA

- Supporting roughly 5,000 researchers across high-performance computing systems with incident triage
- Handling 30+ tickets weekly, resolving scheduler and accounting issues with a 90%+ resolution rate

COLLEGIATE ACTIVITIES

Guidance, Navigation and Control Team

Aug 2025 – Present

Propulsive Landers @ Georgia Tech

Atlanta, GA

- Applying Klamman Filters and nonlinear MPC in Rust for 6 DOF control of rocket lander and UAV prototypes
- Implementing lossless convexification for faster on-device fuel-optimal flight trajectory replanning in Rust
- Prototyping with Monoprop UAV (drone); Aiming for 50m hop with 1000N rocket by EOY 2026

RoboCup: Autonomous Software Team

Aug 2025 – Present

RoboJackets @ Georgia Tech

Atlanta, GA

- Building multi-agent pathfinding, obstacle avoidance and task allocation in C++ for 6v6 robot soccer

PROJECTS, PRESENTATIONS & HACKATHONS

Evaluating Numerical Solving Methods for Optimal Powered Descent

Mar 2026

AIAA Region II Student Conference

Columbia, SC

- Compared ZOH discretization and pseudospectral parameterization to plan 3 DOF trajectories for a rocket lander
- Minimized error from optimal solution at coarse solver resolutions for fast on-the-fly replanning (sub 10ms)

Holonomic Auto-Alignment using CV Localization | C++, Java, ArUco

Aug 2023 – May 2025

- Applied second-order kinematics and jerk limiting to replicate S-curve profile for discretized holonomic drive
- Fused CV estimates from fiducials with filtered odometry using a state-space model for real-time robot location
- Allowed competition robot to align to any scoring position with < 1cm accuracy regardless of starting state

Foundations of Autonomous Programming | Georgia FIRST Robotics Symposium

Sept 2024

- Presented at statewide symposium on methods for teaching autonomous programming to high school students

TECHNICAL SKILLS

Computer Languages: Python, Java, C, C++, TypeScript, Rust

Frameworks & Libraries: TensorFlow, YOLO, Jupyter, ROS, Matplotlib, OpenCV, ArUco, React